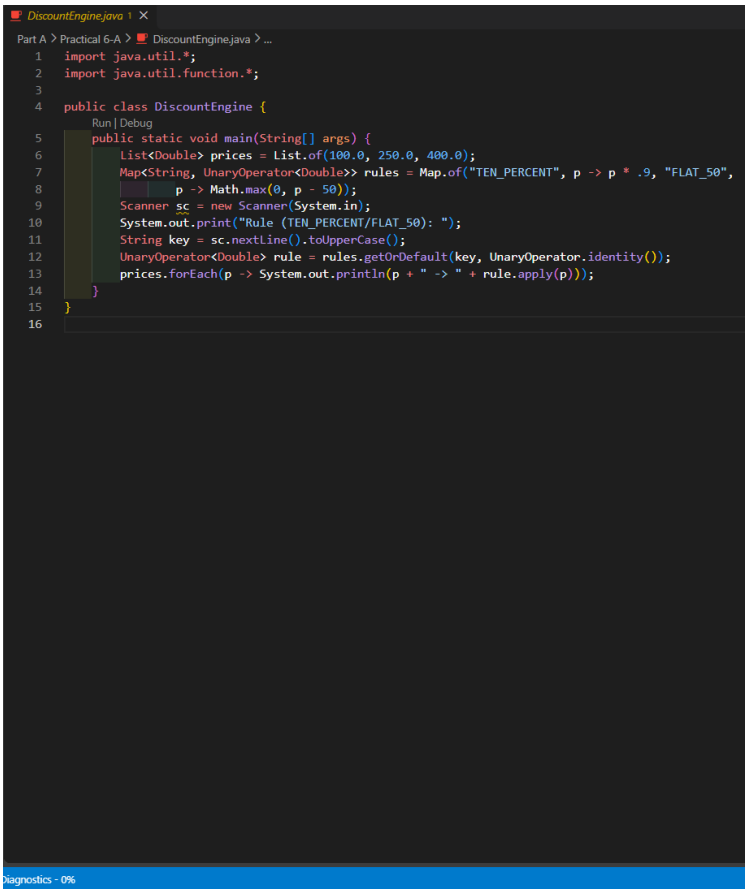
PRACTICAL 6-A

Object Oriented Programming

AIM:

To implement and understand the concepts specified in Practical 6 of the CEUE203 Object Oriented Programming practical list, using Java. The work covers Part A: Remote Control, Notification Senders, Discount Engine and Part B: Interfaces, lambda rule, packages and JAR concepts.

CODE SCREENSHOT:



Code Screenshot 1



```java
@FunctionalInterface
interface Notifier {
    void send(String message);
}

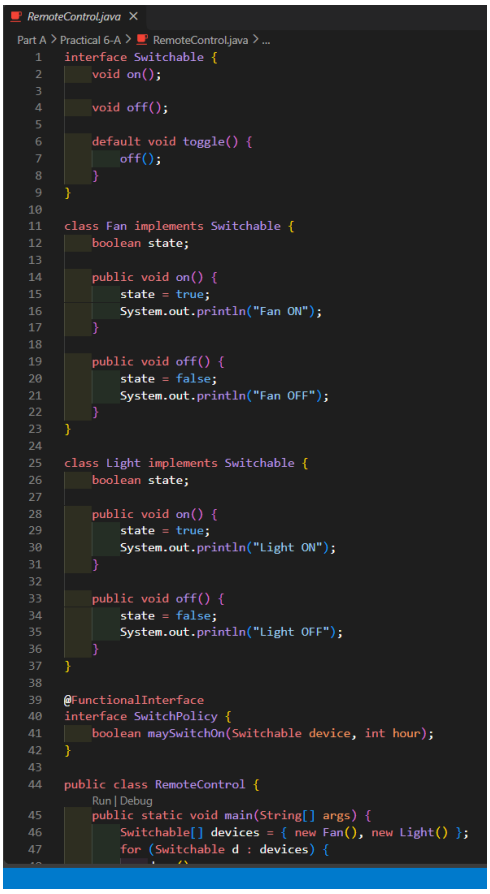
interface Urgent {
}

class EmailNotifier implements Notifier {
    public void send(String m) {
        System.out.println("Email: " + m);
    }
}

class SmsNotifier implements Notifier, Urgent {
    public void send(String m) {
        System.out.println("SMS: " + m);
    }
}

public class NotificationSenders {
    Run | Debug
    public static void main(String[] args) {
        Notifier[] senders = { new EmailNotifier(), new SmsNotifier() };
        for (Notifier n : senders) {
            n.send(message: "System maintenance at 8 PM");
            if (n instanceof Urgent)
                n.send(message: "URGENT: System maintenance at 8 PM");
        }
    }
}
```

Code Screenshot 2



```java
interface Switchable {
    void on();

    void off();

    default void toggle() {
        off();
    }
}

class Fan implements Switchable {
    boolean state;

    public void on() {
        state = true;
        System.out.println("Fan ON");
    }

    public void off() {
        state = false;
        System.out.println("Fan OFF");
    }
}

class Light implements Switchable {
    boolean state;

    public void on() {
        state = true;
        System.out.println("Light ON");
    }

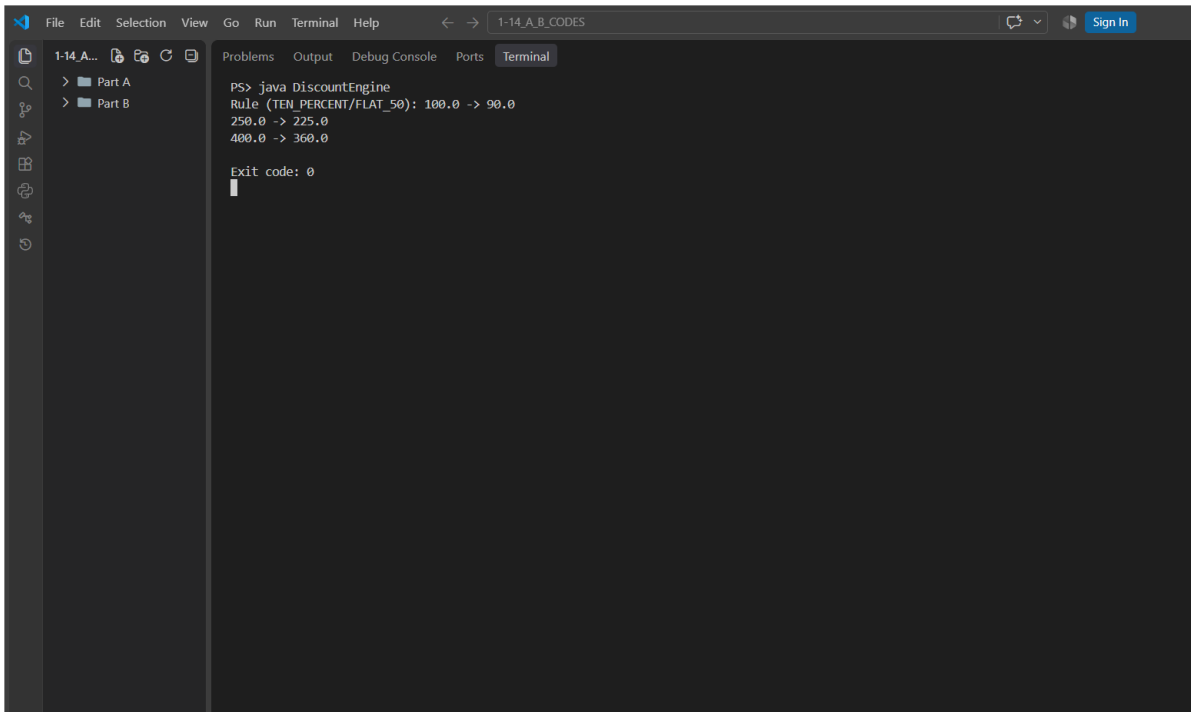
    public void off() {
        state = false;
        System.out.println("Light OFF");
    }
}

@FunctionalInterface
interface SwitchPolicy {
    boolean maySwitchOn(Switchable device, int hour);
}

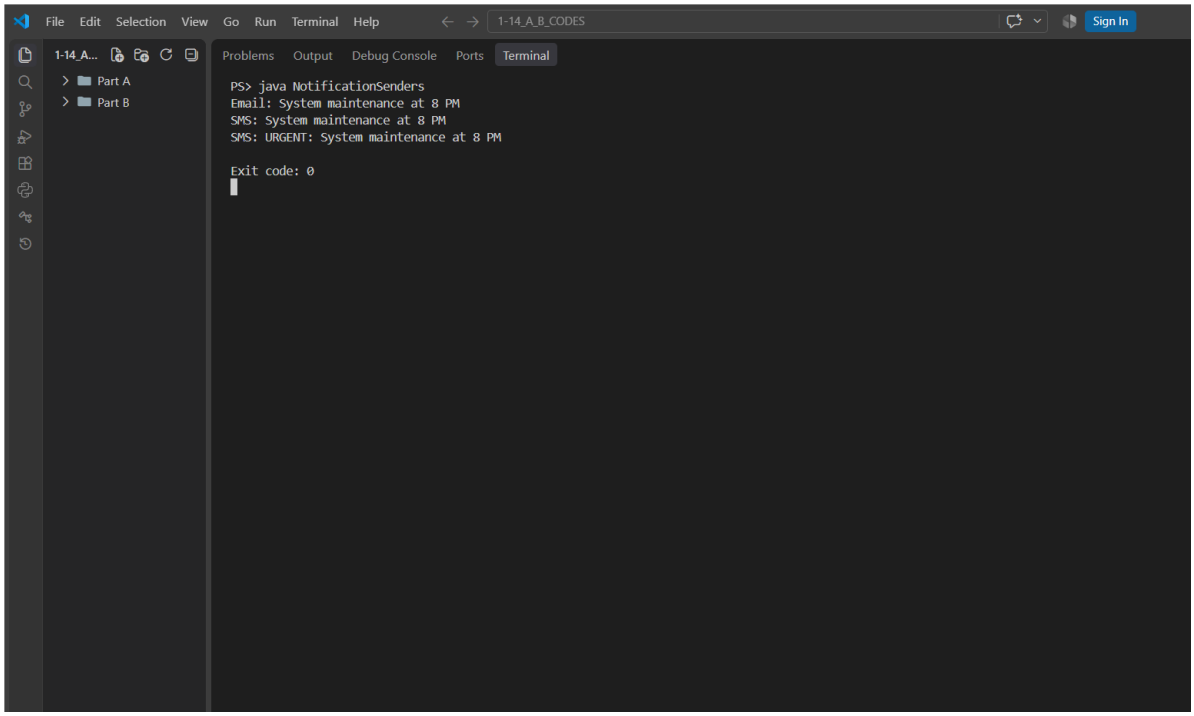
public class RemoteControl {
    Run | Debug
    public static void main(String[] args) {
        Switchable[] devices = { new Fan(), new Light() };
        for (Switchable d : devices) {
```

Code Screenshot 3

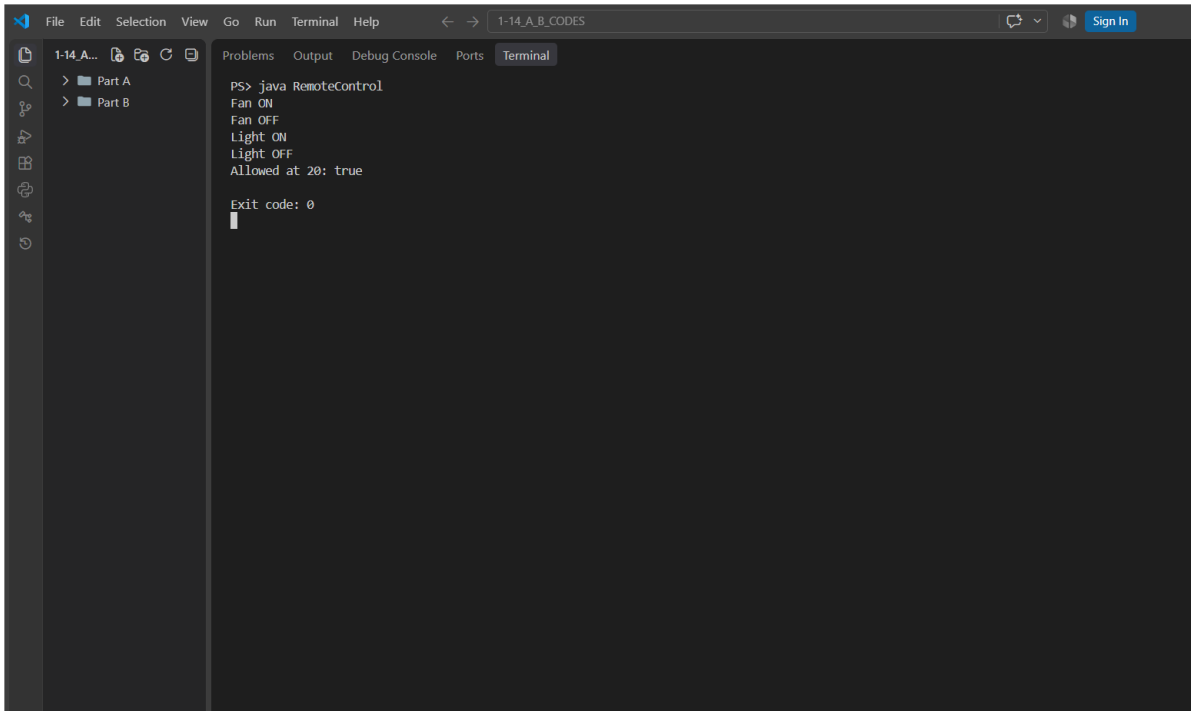
OUTPUT SCREENSHOT:



Output Screenshot 1



Output Screenshot 2



Output Screenshot 3

LEARNING OUTCOMES:

Design programs with interfaces, functional interfaces, lambda expressions, and default methods.